

PROJECT AND TEAM INFORMATION

Project Title

AMBULANCE AND EMERGENCY DISPATCH SYSTEM

Student / Team Information

<i>Team Name:</i> <i>Team #</i>	<i>The Apex Achievers</i>
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Emergency medical services require quick and efficient action because even a small delay can affect a patient's condition. In many situations, people face difficulties in finding an available ambulance, identifying the nearest ambulance, and communicating the emergency details effectively. Traffic conditions and delays in coordination between ambulance services and hospitals can further increase response time. Therefore, there is a need for a technology-based system that can organize the emergency dispatch process and reduce unnecessary delays. The Ambulance and Emergency Dispatch System is motivated by the need to provide a faster, more reliable, and organized method for handling emergency requests. The system focuses on locating available ambulances, assigning a suitable nearby ambulance, tracking its movement, and improving communication with hospitals.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Traditional ambulance systems generally depend on manual emergency calls and manual ambulance allocation. Such systems may have limited vehicle tracking and can experience communication delays, particularly when the number of emergency requests is high. The proposed system improves this process by incorporating GPS-based ambulance location, assisted ambulance assignment, real-time tracking, emergency status updates, and hospital coordination. These features can provide better visibility of ambulance availability and help dispatchers make faster decisions.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

Project Goal To develop an efficient and reliable Ambulance and Emergency Dispatch System. To reduce the time required to respond to emergency requests. To identify and assign the nearest available ambulance to the patient. To use GPS-based location tracking for monitoring ambulances. To provide real-time updates about ambulance availability and emergency status. To improve communication and coordination between ambulance services and hospitals. To maintain emergency records for better management and future reference. To reduce communication errors and improve the overall efficiency of emergency services.

Project Milestones
Requirement Analysis – Study the existing ambulance dispatch process and identify its major problems.
System Design – Design the system architecture involving the emergency caller, dispatcher, ambulance, hospital, server, and database.
Emergency Request Module – Develop a system for receiving and managing emergency requests.
Ambulance Management – Maintain information about ambulance availability and status.
GPS Tracking – Implement location tracking to identify ambulance positions.
Ambulance Assignment – Develop functionality to select and assign the nearest suitable available ambulance.
Dispatch and Tracking – Implement ambulance dispatch and real-time status tracking.
Hospital Coordination – Add hospital information and improve communication between ambulances and hospitals.
Notifications and Records – Implement emergency notifications and maintain emergency history.

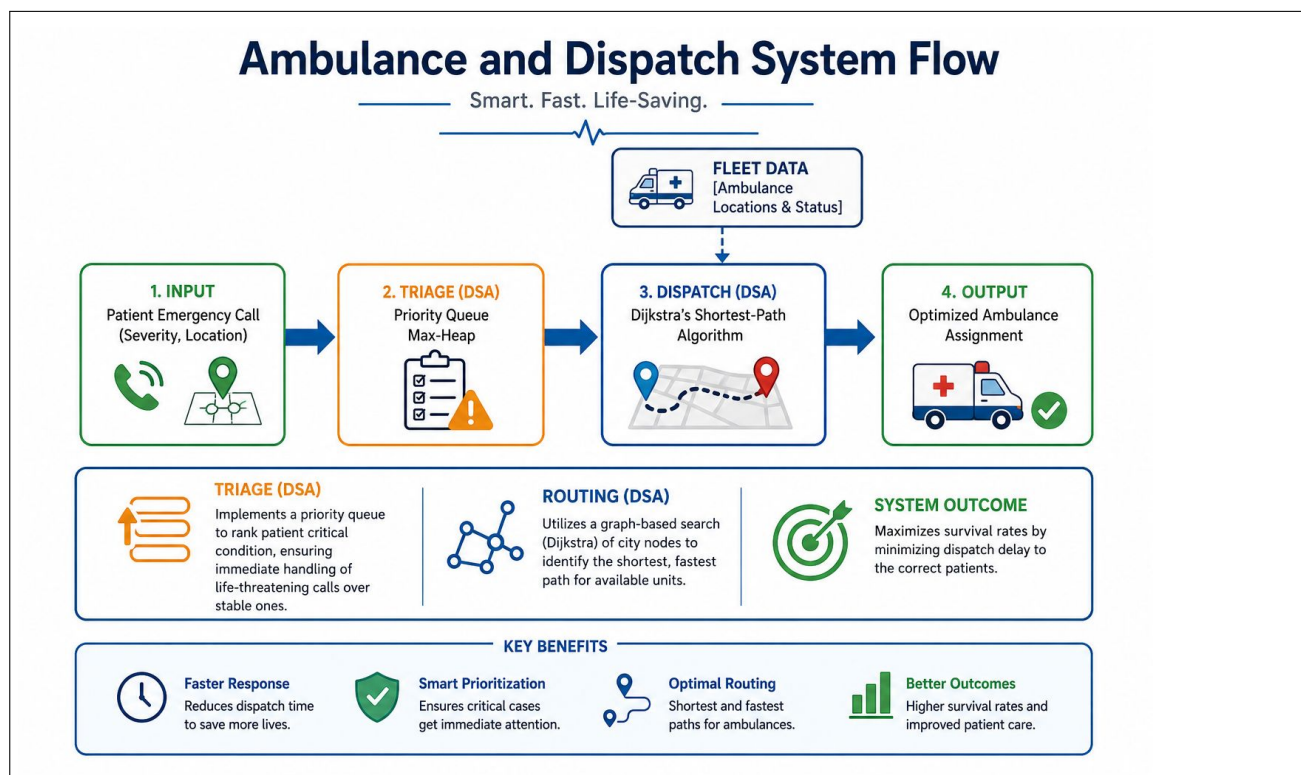
Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

The project will follow a systematic approach to design and develop an Ambulance and Emergency Dispatch System that can manage emergency requests efficiently. First, the requirements of the system will be identified by studying the major problems in the existing ambulance dispatch process, such as delays in ambulance allocation, difficulty in identifying nearby ambulances, and communication problems. The system will maintain ambulance availability information and use GPS-based location services to identify available ambulances. Based on the patient's location and ambulance availability, the system will assist the dispatcher in selecting and assigning the nearest suitable ambulance. After dispatch, the ambulance status and movement can be monitored through real-time tracking. The system will also provide emergency status updates, hospital information, and notifications to improve coordination between the ambulance service and hospital. Emergency records will be stored in a database for future reference and management.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

The project will result in a functional Ambulance and Emergency Dispatch System designed to make emergency response faster, organized, and reliable. The major outcomes and deliverables of the project are:

Emergency Request Management: A system for receiving, recording, and managing emergency requests along with patient location and priority.

Ambulance Management: A module to maintain ambulance availability, status, and basic vehicle information.

Nearest Ambulance Assignment: A feature to identify and assign the nearest suitable available ambulance to an emergency request.

GPS-Based Tracking: Real-time location tracking of the dispatched ambulance to monitor its movement and status.

Dispatch Management: A centralized system that helps dispatchers manage emergency requests and ambulance allocation efficiently.

Hospital Coordination: Hospital information and communication facilities to support patient transportation and emergency coordination.

Notification System: Emergency alerts and status notifications for callers, dispatchers, ambulance drivers, and hospitals.

Emergency Records: A database for storing emergency requests, ambulance trips, response details, and historical records.

User-Friendly Interface: A web/mobile interface that allows different users to access relevant system functions.

Testing and Documentation: Complete system testing along with technical documentation and user guidelines.

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

The system assumes that ambulances have API-enabled devices and their location can be obtained when required. It is assumed that ambulance availability and status information is updated regularly. The system assumes that the patient's location provided during an emergency request is reasonably accurate.

It is assumed that internet/mobile network connectivity is available for communication and tracking.

Hospital information is assumed to be available and maintained in the system database.

It is assumed that dispatchers and ambulance staff have basic knowledge of using the system.

The system assumes that stored emergency data is accessible only to authorized users.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

C++ Reference, Standard Template Library (STL) — vectors, priority queues, maps, and graph-related implementations. GeeksforGeeks, Data Structures and Algorithms, including Graphs, BFS, DFS, and shortest-path algorithms. Robert Lafore, Data Structures and Algorithms in C++, Sams Publishing.